



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECR1013 DIGITAL LOGIC

ELEVATOR CONTROLLER SYSTEM

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DEDICATION & ACKNOWLEDGEMENT

This project is dedicated to our lecturer, Dr Zuriahati Mohd Yunos. We want to give our sincere appreciation to Dr Zuriahati for her guidance in Digital Logic, enabling us to successfully complete this project. Not only did we have the opportunity to learn about digital components and information, but also its real-life applications, which provide us with a holistic understanding toward this course.

In addition, we extend a special thanks to our committed group members. The collaborative effort of our team has been the bedrock of our achievements. Each member brought a unique set of skills and perspectives, fostering a dynamic and creative environment.

This collaboration not only inspired us but also honed our thinking, problem-solving, and teamwork skills. Thank you, Dr. Zuriahati, and our team, for an exceptional learning journey.

TABLE OF CONTENTS

1.0 Objective	1
2.0 Background	1
2.1 Combinational Circuit Components	1
2.2 Sequential Circuit Components	1
2.3 Extra Features	1
3.0 Problem Statement	2
4.0 Suggested Solution	2
4.1 Block Diagram	3
4.2 State Diagram	4
4.3 Next Step and Flip Flop Transition Table	5
4.4 Karnaugh Map (K-Map)	6
5.0 Implementation	7
5.1 Input Switch	7
5.2 Decoder	8
5.3 Clock Disabler	9
5.4 D Flip-flop	9
5.5 3-bit Comparator	10
5.6 7-segment Display	10
6.0 Full DEEDS Circuit	11
7.0 Conclusion	12
8.0 References	12
9.0 Appendices	13

1.0 Objective

The objective of this project is to:

- 1) Design and implement a secure password-protected lift system.
- 2) Incorporate combinational circuit components.
- 3) Utilise sequential circuit elements.
- 4) Validate the circuit design and functionality using the Digital Electronics Deeds Simulator.
- 5) Showcase the application of digital electronics principles in the context of lift.

2.0 Background

In this project, we aim to implement a secured and efficient password-protected lift system. This innovative system requires users to input both their desired floor number and 4 of 2-bit numerical passwords for access. The full circuit is drawn using Digital Electronics Deeds Simulator and includes combinational circuit and sequential circuits.

2.1 Combinational circuit components

- 3-bit comparator
- 4-bit decoder
- Basic gates
- Switches
- LEDs
- Seven segment display

2.2 Sequential circuit components

- Clock disabler
- Clock enabler
- D-flip flops

2.3 Extra Features

- 4-digit password system

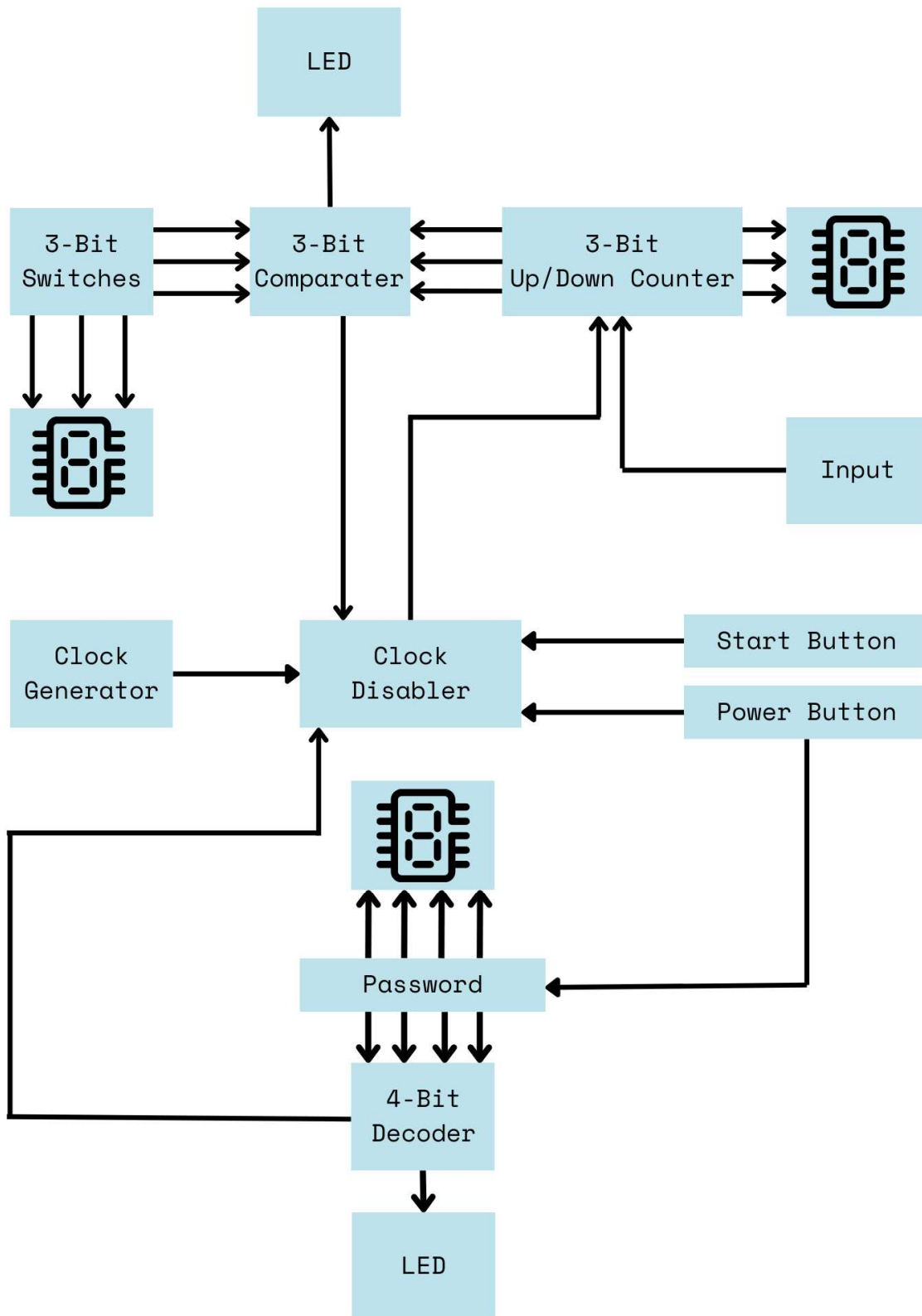
3.0 Problem Statement

In many lifts today, there is a lack of strong security system. It will cause security problems that people will use them without permission. The problem is that existing lifts lack of ways to check if someone is authorised to use them. To solve this problem, we will create a new lift system, by using Digital Electronics Deeds Simulator, that requires a password for access. This new system will use advanced electronic components to make sure only the authorised persons can use it. It will make the lifts safer and more secure by addressing the current problems with unauthorised access.

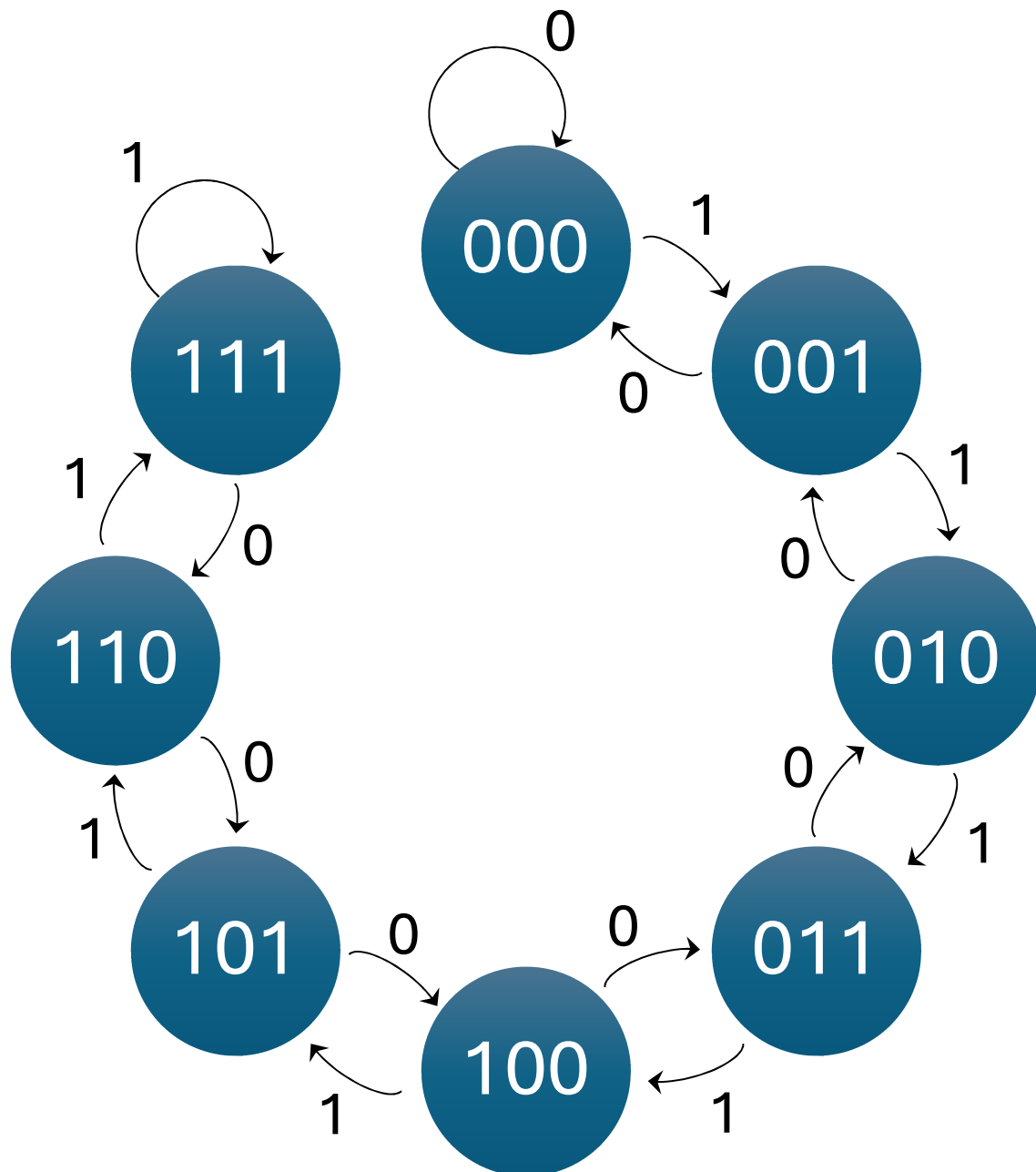
4.0 Suggested Solution

To address the security problem, we propose the implementation of a 4-digit password-protected solution. The suggested solution involves the integration of advanced electronic components, including a 3-bit comparator, 4-bit decoder, basic gates, and D-flip flops. Initially, users are required to enter password to gain access to the lift. The lift will only be activated after the user had entered the correct password in the password reader. Following successful authentication and the LEDs light up, users can change the switch to 1 (indicate going up) and have the option to specify their desired floor level. The lift operation will stop upon reaching the floor level selected by the user. The user can then change the switch to 0 (indicate going down) and have the option to specify their desired floor level. The suggested solution will provide a reliable and efficient authentication mechanism and ensure only authorised users can operate the lift. The proposed system will be design using the Digital Electronics Deeds Simulator.

4.1 Block Diagram



4.2 State Diagram



4.3 Next Step and Flip Flop Transition Table

Input, X	Present State			Next State			D Flip-flop		
	Q ₂	Q ₁	Q ₀	Q ₂₊	Q ₁₊	Q ₀₊	D ₂	D ₁	D ₀
0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	1
0	0	1	1	0	1	0	0	1	0
0	1	0	0	0	1	1	0	1	1
0	1	0	1	1	0	0	1	0	0
0	1	1	0	1	0	1	1	0	1
0	1	1	1	1	1	0	1	1	0
1	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	0
1	0	1	0	0	1	1	0	1	1
1	0	1	1	1	0	0	1	0	0
1	1	0	0	1	0	1	1	0	1
1	1	0	1	1	1	0	1	1	0
1	1	1	0	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1

4.4 Karnaugh Map (K-Map)

$\begin{array}{c} Q_1Q_0 \\ \backslash \\ XQ_2 \end{array}$	00	01	11	10
00	0	0	0	1
01	1	0	0	1
11	1	0	1	1
10	1	0	0	1

$$D_0 = Q_1\bar{Q}_0 + Q_1\bar{Q}_0 + X\bar{Q}_0 + XQ_2Q_1$$

$\begin{array}{c} Q_1Q_0 \\ \backslash \\ XQ_2 \end{array}$	00	01	11	10
00	0	0	1	0
01	1	0	1	0
11	0	1	1	1
10	0	1	0	1

$$D_1 = \bar{X}Q_2\bar{Q}_1\bar{Q}_0 + \bar{X}Q_1Q_0 + X\bar{Q}_1Q_0 + XQ_2Q_1 + XQ_1\bar{Q}_0$$

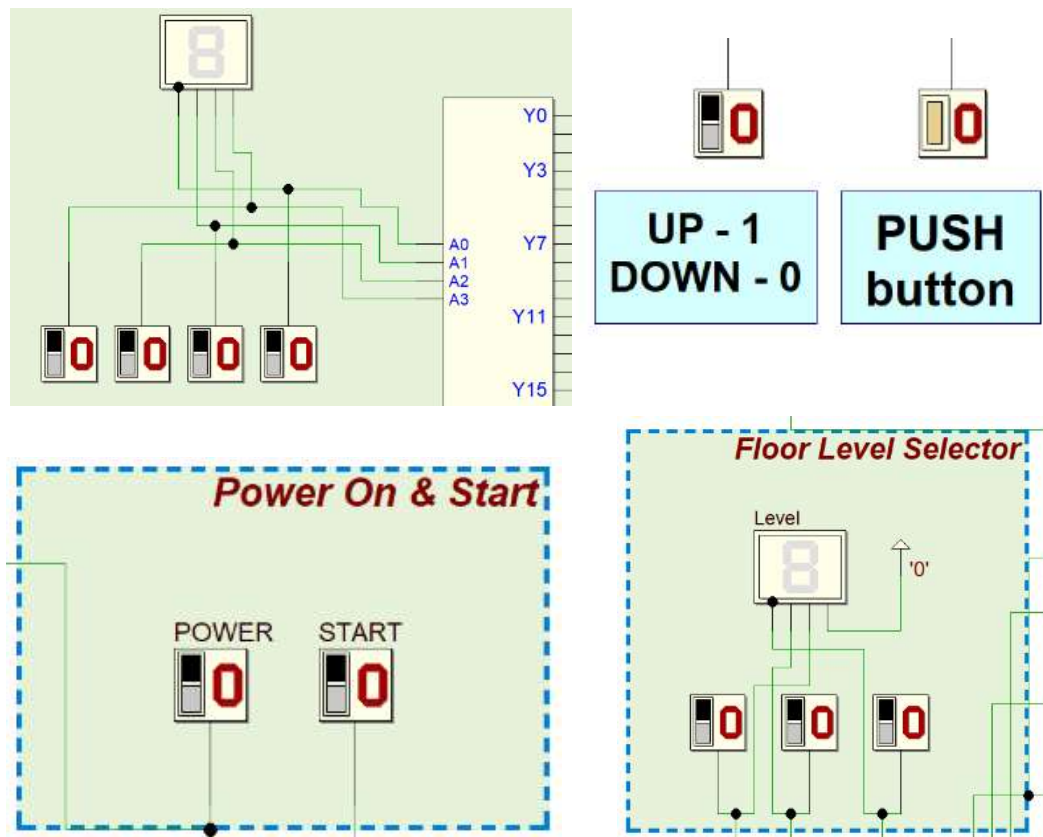
$\begin{array}{c} Q_1Q_0 \\ \backslash \\ XQ_2 \end{array}$	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	1	1	1	1
10	0	0	1	0

$$D_2 = XQ_2 + Q_2Q_1 + Q_2Q_0 + XQ_1Q_0$$

5.0 Implementation

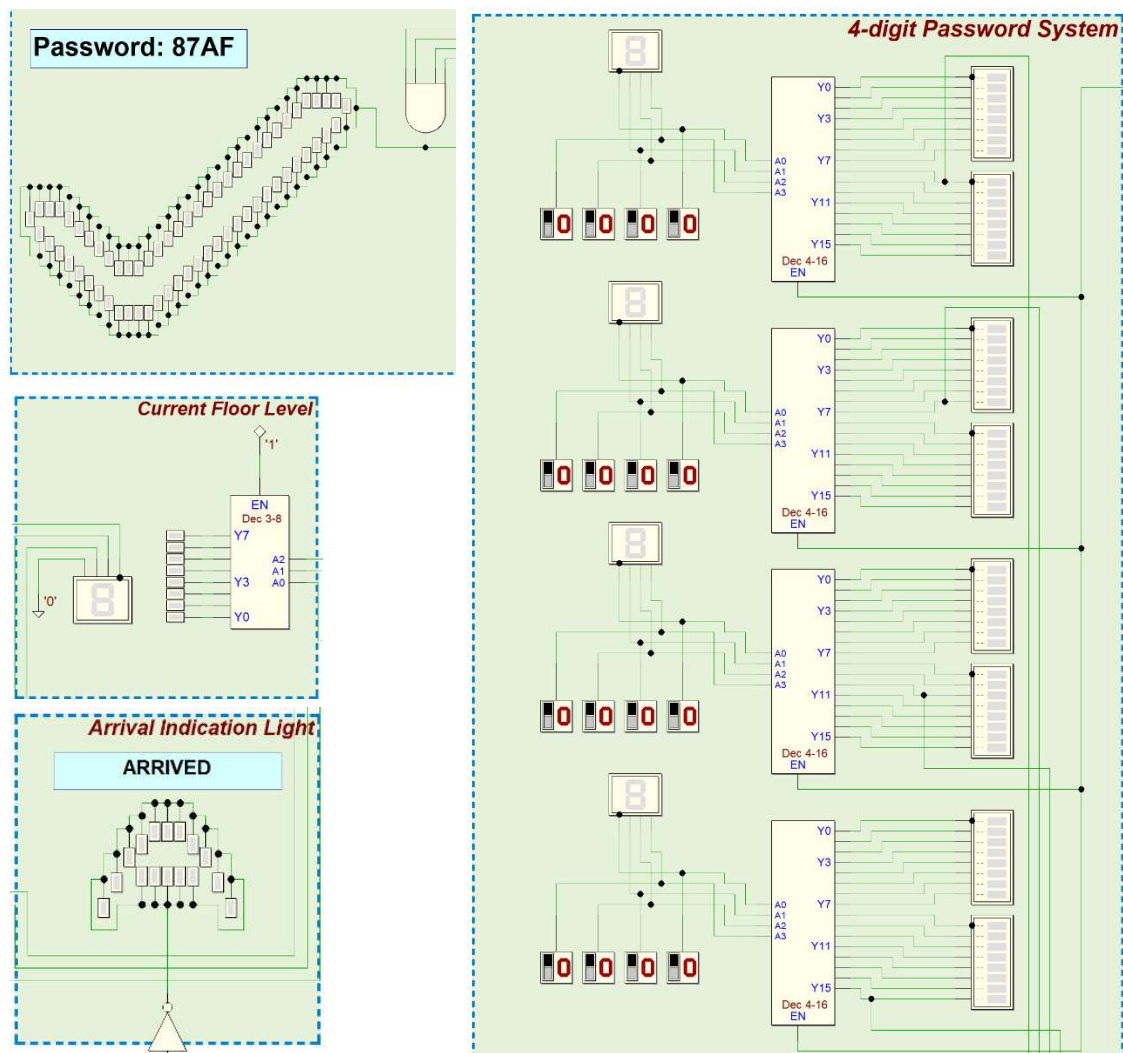
5.1 Input Switch

Input switches allow user to interact with the lift and input relevant information. Specifically, there are 3 input switches for the user to select the desired floor number. The design of the switches represents a single bit respectively, which are Input 1, Input 2, and Input 3. Input 1 represents Most Significant Bit (MSB), whilst Input 3 represents Least Significant Bit (LSB). The user could select the floor level ranging from 0 to 7. The output of the input switch is connected to a 7-segment display to show the user's selected floor number. There are also 16 input switches for the user to input the 4-digit password connected to the 4-bit decoder to decode the set of binary passwords. Besides, there are also 2 input switches to power on, and to start the lift. The lift will only operate if the users power on and start the lift. Additionally, there is an input switch for the user change the flow from going up or going down by selecting 1 (going up floor) or 0 (going down floor).



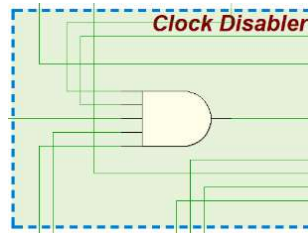
5.2 Decoder

We aim to establish a secure 4-digit password system. In this project, we implement four 4-bit decoders to decode the set of binaries. Each decoder represents each digit of the password. The enable pin of the password reader is connected to the power button of the lift. The password reader will be automatically switched on and activated whilst the lift is powered on. The output of each decoder is connected to two LED arrays. The arrays will light up when the user has selected the input value ranging from 0 to F for each decoder. The output of the decoders is connected to an AND gate, and the AND gate is connected to the tick-icon LED. Once the LED light up, it shows that the user had entered the correct password, enabling the user to proceed to the next step.



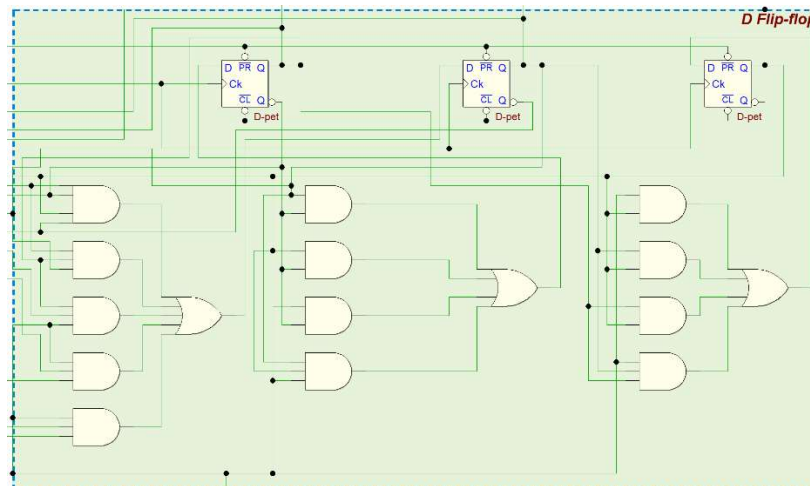
5.3 Clock Disabler

We implement the Clock Disabler using a 5-input AND gate. This 5-input AND gate takes inputs from the push button, a signal from the power-on switch, a signal from the start switch, a signal from the password reader, and a signal from the comparator. The Clock Disabler will only become operational when all four inputs are in the high state (1). The output of the 5-input AND gate is then linked to the clock input of the D flip-flops. The 5-input AND gate is used to start and stop the operation of the lift when the user's selected floor number had arrived.



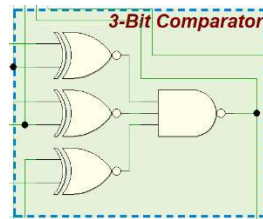
5.4 D Flip-flop

We implement 4-bit D flip-flop in this lift system project. The counter operates based on the clock generator (PUSH button) which determine whether the lift should continue to operate or stop operation. It will start counting if the D input is connected to high state (1) from the clock disabler, and vice versa. Upon correct password entry by the user, the AND gate in the password site will implement the active high (1) input. In the event of a wrong password, the counter remains inactive. The counting process will stop either when the clock pulse is no longer receiving or when the user-specified count is reached.



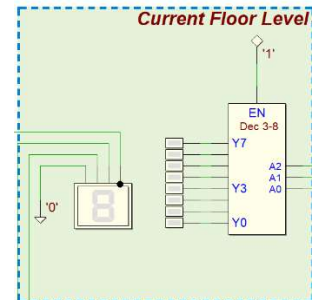
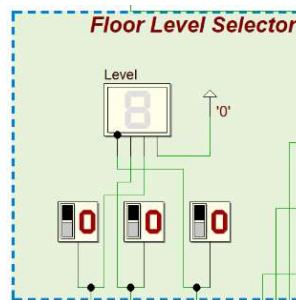
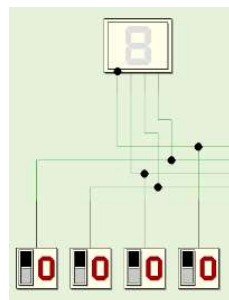
5.5 3-bit Comparator

We implement a NAND gate and three 2-input XOR gates, as the comparator in this project. These components are used to compare value from 2 sources: the input switches, which are used to input the floor level number the user would like to reach, and the current floor level number. The first XOR gate compares the least significant bit (LSB) of the 2 sources. If current floor level number and the user's desired floor level number are the same, the output will be 0, and it is sent to the NOT gate and converted into 1. The same principle can be applied for the second and the third XOR gate. Then, the signal from all three gates will be sent to a NAND gate to convert it into the opposite signal. Therefore, when the input of the NAND gate is received the high input (1) given from input switches and the counter, it will send an output 0 to the clock enabler to stop the lift system.

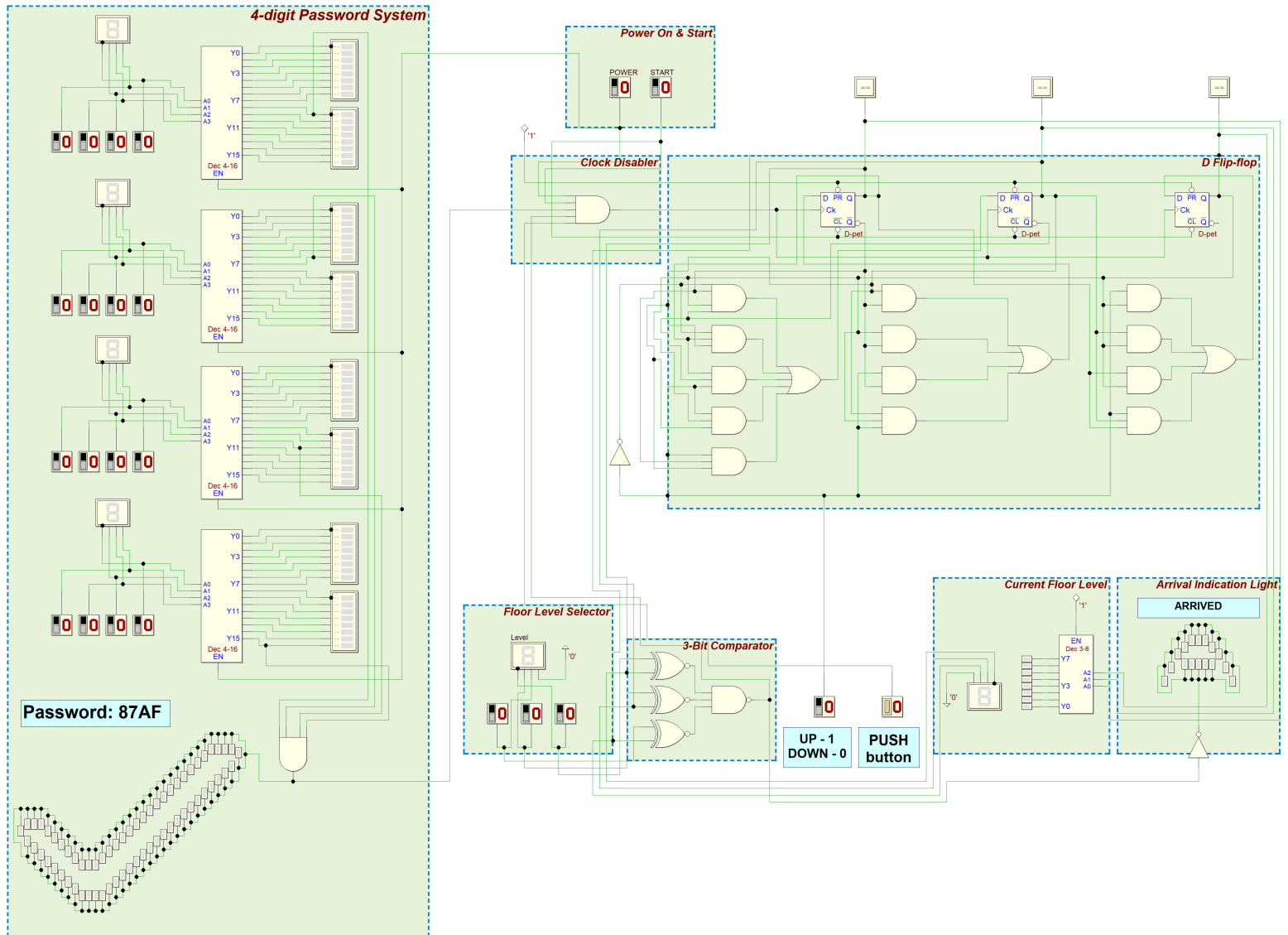


5.6 7-segment Display

The 7-Segment display displays the number based on the BCD value by converting the BCD code received into decimal numbers and letters. This component will display the decimal numbers 0 to 9 and letters from A to F by turning up the proper light of the 7-segment display. The password reader, floor level selector, and current floor level are connected to a 7-segment display. Since the password number ranging from 0 to 15, this component will display the number ranging from 0 to F. The floor level selector will display the floor level number selected by the user, from 0 to 8, and the current floor level will display the current floor of the user.



6.0 Full Digital Electronics Deeds Circuit



7.0 Conclusion

To conclude, the project has a simple, user-friendly design. The circuit mainly consists of four 4-bit decoders, the input switches X, seven segment displays, LEDs for outputs, three 2-bit comparator. The password for security was a great addition to our circuit and was connected so that none of the other components works when it is wrong. Another great option we implemented was the properties. One of the issues we encountered was implementing a 3-bit up/down counter that has saturated feature. In future projects, we are hoping to implement even more complex circuits with more exciting options, and we are looking forward to using our knowledge in related courses and applying it to real-life situations.

8.0 References

- [1] Floyd, T.L. (2018) *Digital Fundamentals*. Uttar Pradesh, India: Pearson India Education Services Pvt. Ltd.
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- [5] Widmer, N.S., Moss, G.L. and Tocci, R.J. (2017) *Digital Systems: Principles and applications*. Boston: Pearson.

9.0 Appendices

Name		Lau Yan Kai	Elijah She Yu Sheng	Chew Chiu Xian	Brendan Chia Yan Fei
Task	Designing	Full Circuit using Digital Electronics Simulator			
		Cover page	Conclusion		
		Video Editing		Script preparation	
		State diagram, Next-state and D FF Table, K-map	Block diagram		
	Report Writing	Problem and Solution, System Implementation	Background	Acknowledgement and Dedication, Conclusion, Objective	Requirements and Components

